

- 2** A program stores data in a 1D array of records, `HashTable`. The array has space for 200 records. Each data item is stored in a specific index of the array that is calculated using an algorithm. The index is calculated from the key field using the formula: $\text{Key} \bmod 200$

If two key fields generate the same index, there is a collision. Any records that have a collision are stored in a second array, `Spare`. This array has space for 100 records. When a collision is detected, the record is stored in the next free space in `Spare`.

- (a)** The pseudocode record format is:

```
TYPE NewRecord

    DECLARE Key : INTEGER

    DECLARE Item1 : INTEGER

    DECLARE Item2 : INTEGER

ENDTYPE
```

Write program code to declare the record type `NewRecord`

If your chosen programming language does **not** support record formats, a class can be used instead.

Save your program as **Question2_J25**.

Copy and paste the program code into part **2(a)** in the evidence document.

[2]

- (b) (i)** Write program code to declare the global arrays `HashTable` and `Spare`

Save your program.

Copy and paste the program code into part **2(b)(i)** in the evidence document.

[1]

- (ii)** An empty record has the integer `-1` stored in each field.

The procedure `Initialise()` stores an empty record in each element in `HashTable` and `Spare`

Write program code for `Initialise()`

Save your program.

Copy and paste the program code into part **2(b)(ii)** in the evidence document.

[2]